THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET B

Chapter 2: **Electricity: Circuits and Their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The two terminals of a cell are: 1

(a) red and blue only **(b)** north and south
(c) positive (+) and negative (−) **(d)** hot and cold

2. A bulb does not glow when the switch is: 1

(a) closed (ON) **(b)** connected **(c)** in the circuit **(d)** open (OFF)

3. A representation of a circuit using symbols is a: 1

(a) photograph **(b)** chart **(c)** map **(d)** circuit diagram

4. Graphite in a pencil is: 1

(a) a solvent **(b)** an acid **(c)** an insulator **(d)** a conductor

5. A bulb connected to a cell glows because: 1

(a) wires glow **(b)** the cell shines
(c) the switch emits light **(d)** current heats the filament

6. When you press an electric bell, the circuit gets: 1

(a) open **(b)** closed **(c)** cut **(d)** short only

7. An electrical component that protects circuits by melting is a: 1

(a) switch **(b)** cell **(c)** fuse **(d)** bulb

8. Pure distilled water is: 1

(a) a poor conductor **(b)** an acid **(c)** a good conductor **(d)** a metal

9. The symbol for an open switch shows: 1

(a) a zigzag **(b)** a closed line **(c)** a circle **(d)** a gap

10. The unit of electric current is: 1

(a) gram **(b)** metre **(c)** ampere **(d)** second

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. What is a filament? What happens when current flows through it? 2

12. Draw the symbol of a cell and a switch (describe). 2

13. Why are electrical wires coated with plastic? 2

14. What is a circuit diagram? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the difference between AC and DC with examples. 3

16. What are the safety rules when working with electricity? 3

17. Why does a torch bulb not glow if the cell is inserted wrongly? 3

OR

What is the role of the switch in a circuit? Give an everyday example. 3

18. What happens to the lamp when two cells are connected in the correct order versus the wrong order? 3

OR

Explain why salt water conducts electricity but distilled water does not. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Describe the working of a torch. 5
(ii) Name its parts.
(iii) Why does the bulb glow?

OR

(i) Explain AC and DC. 5
(ii) State their sources.
(iii) Why do we not use AC in experiments in class?

20. **Case study:** A student tests lemon juice, sugar solution, salt solution and distilled water in a tester. The bulb glows for lemon and salt only. 5
(i) Conclusion?
(ii) Why not for sugar?
(iii) Why not distilled water?
(iv) Why add salt to water?

OR

(i) What are the effects of electric current? 5
(ii) Which one is seen in a bulb?
(iii) Which in a heater?

END OF QUESTION PAPER — Total: 40 Marks

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